# PATEL MOHAMMED FAIZAN

## Mechanical Engineering Postgraduate

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## SUMMARY

Motivated Mechanical Engineering postgraduate specializing in Machine Design with strong proficiency in CAD, FEA, and product design. Seeking to apply technical expertise, deliver high-quality engineering solutions, and continuously enhance skills.

## EXPERIENCE

### Internship

**Modelyzelabs AI, Bengalore**

*Jan 2026 – Apr 2026*

Mechanical Engineering graduate working as DPC at ModelyzeLabs with experience in data processing and documentation. Knowledge of MEP systems including HVAC, plumbing, and fire protection. Skilled in ANSYS, Finite Element Analysis, and engineering analysis.

### Internship

**Bosch, Bengalure**

*Dec 2025 – Jan 2026*

Motivated engineering graduate with hands-on internship experience supporting design, analysis, and testing activities. Assisted senior engineers in product development, performed basic simulations, and contributed to technical documentation. Skilled in engineering fundamentals, data analysis and cross-functional collaboration.

### Intern, Wheel & Axle Manufacturing

**Rail Wheel Factory, Bangalore**

*08/2022 – 09/2022*

- Gained practical exposure to large-scale manufacturing processes in wheel and axle production
- Observed and learned about quality control and machining operations in railway components

## PROJECTS

### M.Tech Thesis – Machine Learning-Based Prediction of Fatigue Crack Growth Rate and Fatigue Life in SAE 1020 Steel

**Published at SICE 2026 – Structural Integrity Procedia, Elsevier**

*2025 – 2026*

- Developed a dual-output machine learning framework predicting fatigue crack growth rate (da/dN) and remaining fatigue life in SAE 1020 steel across stress ratios R = 0.1–0.8
- Trained and compared five models – Linear Regression, Random Forest, Gradient Boosting, a residual deep network (FCGR-Net) and a Physics-Informed Neural Network – on a 2,312-sample dataset of five specimen geometries
- Validated predictions against the Paris–Erdogan analytical solution and an ANSYS Workbench SMART crack-growth model of CTS01, with SIF within 0.73% of the analytical result

### Automatic Fire Protection System

*11/2022 – 06/2023*

- Designed an automatic fire detection and suppression system
- Integrated sprinklers, alarms, and suppression systems
- Ensured compliance with safety codes and performed risk assessments

### Charging of EV Batteries using Wind Energy

*06/2021 – 11/2022*

- Designed a wind energy-based charging system for EV batteries
- Integrated energy storage systems for stable power supply
- Contributed to sustainable transportation by reducing reliance on fossil fuels

### Failure Analysis of Bracket using ANSYS Workbench

*07/2025 – 08/2025*

- Performed static and fatigue failure analysis of a steel bracket using ANSYS Workbench
- Evaluated stress distribution, deformation, fatigue life, and factor of safety under cyclic loading
- Conducted parameterization studies by varying bracket thickness and width to propose optimized design solutions

## DECLARATION

*I hereby declare that the information provided above is true and correct to the best of my knowledge and belief*

## KEY ACHIEVEMENTS

**◆Increased Charging Efficiency**

Achieved 25% efficiency increase in EV battery charging system using renewable energy.

**◆Optimized Bracket Design**

Reduced failure rates by 15% through optimized design solutions in ANSYS Workbench.

**◆Ensured Safety Compliance**

Ensured compliance with safety codes for fire systems, covering over 10,000 square feet.

**◆Improved Manufacturing Output**

Successfully completed an internship at Rail Wheel Factory, enhancing production by 10%.

## EDUCATION

**M.Tech in Machine Design**

**B.M.S College of Engineering, Bengaluru**

12/2024 -09/2026

**CGPA: 8.9 / 10.0**

**B.E Mechanical Engineering**

**G.M. Institute of Technology, Davanagere**

09/2019 – 07/2023

**CGPA: 8.2 / 10.0**

## SKILLS

**CAE-Software**

ANSYS Workbench (Static Structural, Fatigue, SMART Crack Growth), Abaqus

**CAD-Software**

Solid Edge, SolidWorks

**Programming Knowledge**

Python, Java

**Domain**

Fracture mechanics, fatigue & damage-tolerant design, Paris–Erdogan law, ASTM E647, machine design, GD&T basics

**MS Office**

Word, PowerPoint

## LANGUAGES

**English**

**Hindi**

**Urdu**

**Kannada**

## STRENGTHS

- Excellent analytical and creative skills
- Structured and strong teamwork
- Independent and strong problem-solving skills
- Time Management